



United International University (UIU)

Dept. of Computer Science & Engineering (CSE)

Mid Exam Spring 2026

CSE 2233/CSI 233: Theory of Computation/Theory of Computing

Total Marks: 30

Duration: 1 Hour 30 Minutes

Answer all questions. Figures in the right-hand margin indicates full marks.

Any examinee found adopting unfair means will be expelled from the trimester / program as per UIU disciplinary rules.

1. Design DFAs that accepts the following languages $\Sigma = \{ a, b \}$: 3x3

a) $L = \{ w \mid w \text{ starts with "ab", contains "aab" as a substring, and ends with "bab" } \}$

Example: abbcaabab , abaabcbcbab

b) $L = \{ w \mid \text{total number of "a" in the string is even and } w \text{ ends with "bba" } \}$

Example: abba , babaabba

c) $L = \{ w \mid w \text{ starts with a non-zero even number of "a" and ends with an odd number of "b"} \}$

Example: aaaabbb , aabbbbbb

2. a) Design NFAs that accept the following languages: 3x3

i) $L = \{ w \mid w \text{ starts with "a" and second last letter of the string is "b" or } w \text{ starts with "c" and last letter of the string is "c" } \mid \Sigma = \{ a, b, c \} \}$

Example: ababcabc, cabbac

ii) $L = \{ w \mid \text{length of } w \text{ is a multiple of "2" and every "1" is followed by a "0"} \mid \Sigma = \{ 0, 1 \} \}$

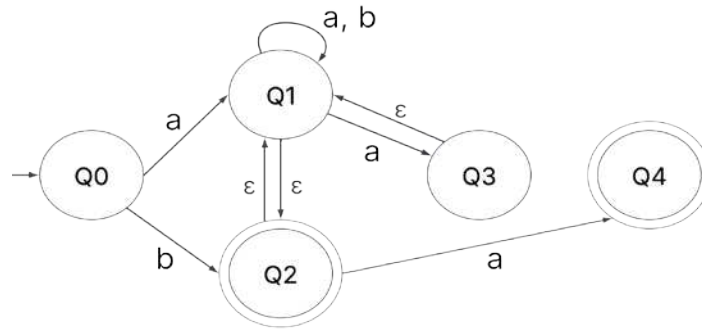
Example: 001010, 101010

b) Design ϵ -NFA that accept the following language

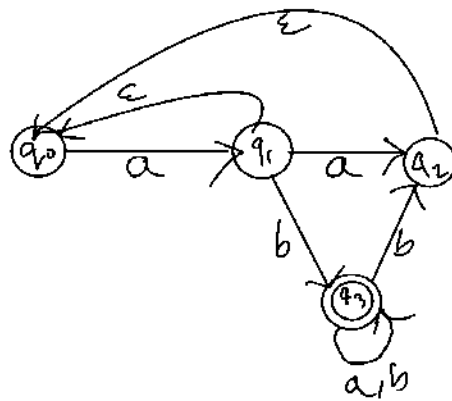
i) $L = \{ w \mid w \text{ starts with "xx" or "xy", contains "yyx" or "zzz" as a substring, and ends with "yx" } \mid \Sigma = \{ x, y, z, \epsilon \} \}$

Example : xxεyyxεεyx

- 3 a) Consider the following ϵ -NFA, and show with the help of ϵ -NFA-tree whether the string “baa” is accepted or not. Here, $\Sigma = \{a, b, \epsilon\}$ 3



4. a) Convert the ϵ -NFA to equivalent DFA. Show only the transition table of the DFA. Here, $\Sigma = \{a, b, \epsilon\}$: 3



- 5 Design Regular Expression for the following languages: 2x3
- $L = \{ w \mid w \text{ starts with an even number of '0's and ends with '112' } \}, \Sigma = \{0,1,2\}$
 - $L = \{ w \mid w \text{ has no two consecutive 'b's' } \}, \Sigma = \{a,b\}$
 - $L = \{ w \mid w \text{ starts and ends with the same letter while the last two letters of } w \text{ are different } \}, \Sigma = \{a,b\}$

